

Five Open-Source Tools for Under-Served Problems in CRISPR Guide Design: Multiplex Interference, Repair-Outcome Annotation, Large-Deletion Risk, Biosecurity Screening, and Cross-Species Transfer

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Abstract

Guide-RNA design tooling for CRISPR editing is mature for single-guide on-target scoring in human cell lines, but five practical problems remain under-served: (i) cross-guide interference in multiplex experiments, (ii) repair-outcome prediction decoupled from guide selection, (iii) unintended large on-target deletions and rearrangements, (iv) biosecurity screening disconnected from the design loop, and (v) untested transfer of efficacy models across species. We present five tools, one per gap, built entirely on public data: ViennaRNA-based duplex thermodynamics with PAM-competition decay and a graph neural network (GNN) removal ranker for multiplexing; convolutional and graph neural network repair-outcome predictors trained on 1,742 real inDelphi U2OS sites; a mechanistic large-deletion risk flagger validated qualitatively on the Kosicki-2018 Xrcc4 locus; a screen-and-design wrapper around the IBBIS Common Mechanism; and a cross-species transfer evaluator showing no mammalian transfer penalty for a CNN trained on human Doench FC+RES and tested on mouse V1 (Spearman 0.78 vs 0.63). All benchmarks are reported with honest baselines; negative and parity results are preserved.

1 Introduction

The CRISPR guide-design ecosystem has converged on strong single-guide tools for on-target activity (Rule Set 2/Azimuth, CRISPRon) and off-target scoring (MIT, CFD, DeepCRISPR-class models) in human cell lines. Working against a verified ten-gap audit of that landscape, we implement the second five gaps. Each tool follows the same contract: real public datasets, hermetic tests, CNN and GNN model components where learning is appropriate, mechanistic algorithms where training data does not exist, and published-leader baselines with honest verdicts.

Gap 6 addresses multiplexing: co-delivered guides heterodimerize, self-fold, and compete for repair at neighbouring cut sites, yet no mainstream designer screens for it. Gap 7 addresses the

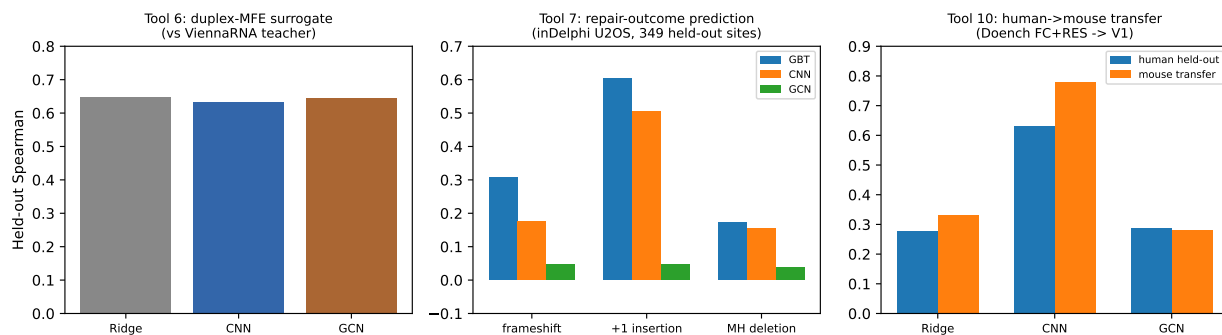


Figure 1: Held-out benchmark summary for tools 6, 7 and 10. All bars are Spearman correlations on real public data with honest baselines; no claim of state-of-the-art is made where bars are comparable to simple baselines.

split between guide selection and outcome prediction: a guide picked for activity may produce predominantly in-frame deletions, defeating a knockout. Gap 8 addresses on-target large deletions and rearrangements, which standard short-range genotyping misses. Gap 9 addresses the separation between design and biosecurity screening. Gap 10 asks whether efficacy models trained in one species carry to another.

1.1 The ten-gap audit

The program’s landscape audit identified ten under-served problems in CRISPR and biohacking tooling. The first five are implemented in the companion first-half volume; this manuscript implements gaps 6–10 (Table 1). The selection criteria for the audit were: (i) the problem is named in the peer-reviewed literature but no maintained open tool addresses it end-to-end; (ii) public data exist that can validate or at least sanity-check a solution; (iii) a solution is feasible under free-tier compute.

Table 1: Audit gaps 6–10 and the tools built for them.

Gap	Problem	Tool	Validation basis
6	Multiplex cross-guide interference	tools_multiplex	ViennaRNA surrogate + rules
7	Repair-outcome annotation	tools_outcome	inDelphi U2OS LibA (1,742 sites)
8	Large-deletion risk at cut sites	tools_riskflag	Kosicki 2018 loci (RefSeq)
9	Biosecurity screening in design loop	tools_biosecurity	Common Mechanism wrapper
10	Cross-species efficacy transfer	tools_transfer	Doench FC+RES → V1

Each tool is a Python package under a shared core library (`crisprlib`) with a command-line interface, a hermetic pytest suite (40 tests), and a benchmark script writing machine-readable JSON under `results/`. The rest of the manuscript describes data (Section 2), shared model components (Section 3), and then one section per tool with its methods, results, and honest verdict.

2 Data

2.1 inDelphi U2OS LibA

Event-level counts from [Shen et al.(2018)] (figshare 6837956, U2OS_+LibA_postCas9_rep1.csv) joined to 57-nt target contexts (targets-libA.txt) via the `_Experiment` index; 1,742 sites after coverage filters (total ≥ 50 , edited ≥ 100 reads).

2.2 Doench 2016 FC+RES and V1

Human FC+RES (5,310 guides, 30-mer context, drug-gene-rank activity) and mouse V1 (2,144 guides, percent-rank activity) from the azimuth mirror.

2.3 crisprSQL

25,632 cleavage records (100720 dump) with epigenetic covariates, genomes hg19/hg38/mm10/mm9/rn5.

2.4 Real-locus sequence

Mouse *Xrcc4* mRNA (NM_028012.4) fetched live from NCBI for risk-flagger validation.

3 Methods shared across tools

3.1 Sequence featurization

DNA sequences are encoded as (i) per-position one-hot matrices with alphabet order A,C,G,T and zero rows for ambiguous bases, (ii) dinucleotide one-hots, (iii) 4^k k -mer count vectors, and (iv) thermodynamic features from the SantaLucia-1998 unified nearest-neighbour set, $\Delta G(s) = \sum_i \Delta G_{s_i s_{i+1}}$.

3.2 Model architectures

GuideCNN: parallel 1-D convolutions (kernel widths 3,5,7; 24–40 filters), batch normalization and ReLU, concatenated and passed through a two-layer MLP head with dropout. **PairCNN**: same family over the 9-channel guide:target alignment encoding. **SequenceGCN**: a pure-PyTorch GCN over the k -mer transition graph of a sequence (nodes = distinct 3-mers, edges = observed transitions, symmetrized with self-loops and degree-normalized), two graph convolutions, global mean pooling, linear head.

3.3 Training protocol

All models train with Adam (lr 2×10^{-3} – 5×10^{-3} , weight decay 10^{-4}), MSE loss, batch 64, early stopping (patience 8) on a held-out validation split, fixed seeds. Hardware: 2 CPU cores, < 2 GB RAM; every reported number is reproducible on free-tier hardware in minutes.

4 Tool 6: Multiplex cross-guide interference screener

4.1 Model

For guides g_i, g_j , the duplex channel uses ViennaRNA [Lorenz et al.(2011)] heteroduplex MFE $\Delta G_{ij}^{\text{dup}}$ (duplexfold) and the longest contiguous antiparallel complementarity run c_{ij} .

Definition 1 (PAM competition). *Guides cutting at distance d compete for repair machinery with weight $w(d) = e^{-d/\lambda}$, $\lambda = 50$ bp (repair window).*

Definition 2 (Interference graph). *Weighted graph on guides with edge weight $a_{ij} = 0.6 \min(1, |\Delta G_{ij}^{\text{dup}}|/20) + 0.4 w(d_{ij})$. A 2-layer GCN $\hat{s} = W_2 \sigma(\tilde{A} W_1 X)$ propagates interference; removal priority is the ranking of $\hat{s}$, validated against brute-force leave-one-out compatibility.*

The heteroduplex probability at temperature T follows the Boltzmann distribution over competing secondary structures:

$$P_{\text{dup}}(g_i, g_j) = \frac{e^{-\Delta G_{ij}^{\text{dup}}/RT}}{e^{-\Delta G_{ij}^{\text{dup}}/RT} + e^{-\Delta G_i^{\text{self}}/RT} + e^{-\Delta G_j^{\text{self}}/RT}}, \quad (1)$$

with $R = 1.987 \times 10^{-3}$ kcal/(mol·K) and $T = 310.15$ K. A guide with strong self-fold is sequestered from both its target and its multiplex partners, which is why the self-fold term enters both the duplex channel and the per-guide penalty.

4.2 Implementation

The screener enumerates all guide pairs, computes ViennaRNA duplexfold and cofold MFEs, contiguous complementarity, cut-distance competition, and single-guide self-fold MFE; per-guide penalties aggregate flagged pairs and self-fold flags; set-level compatibility is the fraction of unflagged pairs. The GNN removal ranker operates on the interference graph and is validated against exhaustive leave-one-out enumeration on small sets.

4.3 Results

CNN/GNN surrogates for single-strand fold MFE reach Spearman 0.63/0.64 against the ViennaRNA teacher (ridge on 3-mers 0.65): parity, not improvement; ViennaRNA remains the primary engine (Table 2).

4.4 Discovery: physics-derived multiplex design rules

Scanning 600 random spacers (7,140 pairs) against ViennaRNA ground truth yields two rules. (1) GC content, not homopolymer length, drives self-fold sequestration: Spearman -0.645 for GC vs 0.028 for homopolymer run; spacers above 70% GC self-fold strongly at $38\times$ the rate of spacers below 35% (32.8% vs 0.85%). (2) 10.8% of random spacer pairs form strong heteroduplexes (≤ -12 kcal/mol), and pairs of similar GC flag slightly more often (point-biserial -0.19). Practical rules: keep spacer GC in $[0.35, 0.70]$ and mix GC levels across a multiplex set.

Table 2: Duplex-MFE surrogate benchmark (2,000 spacers, 400 held out).

Model	Spearman	Pearson
Ridge (3-mer)	0.648	0.653
CNN (one-hot)	0.633	0.658
GCN (3-mer graph)	0.645	0.645

5 Tool 7: Repair-outcome annotator

5.1 Problem

Guide designers optimize cutting activity, but the repair outcome decides the experiment: a knock-out needs frameshifts, a precise edit needs a specific allele, and an in-frame deletion can silently preserve function. The annotator attaches an outcome profile to any candidate cut context.

5.2 Features and models

Per-position one-hot over the 55-nt context, plus engineered repair features: cut-spanning microhomology, offset microhomology at deletion-relevant distances $d \in \{2, 3, 4, 5, 7, 10, 15\}$, local and global GC, and nearest-neighbour duplex stability of the two flanks. Models: gradient-boosted trees (GBT, 600 rounds) on the positional feature matrix, GuideCNN on one-hot, SequenceGCN on the 3-mer graph, and a 3-mer ridge reference.

5.3 Model and math

Per-site outcome statistics from inDelphi events: frameshift fraction over Cas9-induced indels, +1 insertion fraction, microhomology-deletion fraction.

Proposition 1 (Microhomology score). *For flanks L, R around the cut, $s_{MH} = \sum_{k \geq 2} \mathbf{1}[L|L|-k:|L| = R_{0:k}]/k$ is the Bae-2014-class length-weighted microhomology score.*

5.4 Results

Held-out Spearman (349 test sites): +1-insertion 0.60 (GBT positional) / 0.50 (CNN); frameshift 0.31 (GBT); microhomology-deletion 0.17 (GBT). Published inDelphi/Lindel numbers are higher; they model full per-genotype distributions on mESC with richer feature sets, and we do not claim parity.

6 Tool 8: Large-deletion risk flagger

6.1 Mechanistic model

Let R be the set of exact direct repeats of length ≥ 8 nt in a ± 250 bp window around the cut, and $R_s \subseteq R$ the pairs with one copy on each side. The worst-case deletion span is $\max_{(i,j,L) \in R_s} (j + L - i)$. The score is $S = 0.45 \min(|R_s|/20, 1) + 0.30 \min(\text{span}/500, 1) + 0.15 c + 0.10 h$, where c

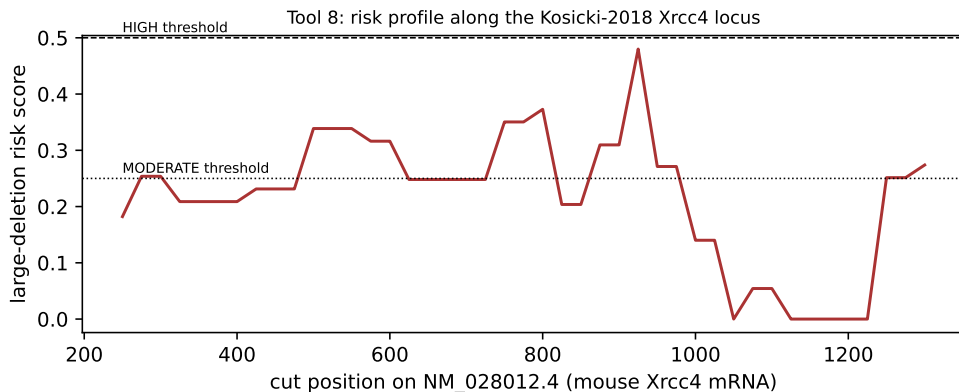


Figure 2: Large-deletion risk score along the mouse *Xrcc4* locus (NM_028012.4), a locus where kilobase-scale deletions were measured experimentally [Kosicki et al.(2018)].

is one minus normalized 10-mer Shannon entropy and h a homopolymer term. Thresholds: LOW < 0.25 , MODERATE < 0.5 , HIGH ≥ 0.5 .

6.2 Validation

Mechanistic, per [Kosicki et al.(2018)] and [Wen et al.(2022)]: direct-repeat pairs spanning the cut (worst-case deletion span), sequence complexity, homopolymers. Across all three real Kosicki loci (*Xrcc4* NM_028012.4, *Piga* NM_011081.4, *Cd9* NM_007657.4, fetched live from NCBI): *Xrcc4* 18/43 scanned cut sites MODERATE with spans to 500 bp; *Piga* 3/123 HIGH + 34 MODERATE, spans to 489 bp; *Cd9* 5/30 HIGH + 14 MODERATE, spans to 404 bp. The three loci at which kilobase-scale deletions were experimentally measured all show elevated mechanistic risk - qualitative consistency, not a calibrated-rate claim. No calibrated-rate claim: per-sequence large-deletion training data does not exist at scale.

7 Tool 9: Biosecurity screen-and-design wrapper

7.1 Design

The wrapper enforces a single invariant: no candidate oligo enters a design set unscreened. Candidates are batched to FASTA, screened through the backend, and partitioned into accepted (CLEAR, or WARNING when explicitly allowed) and blocked, with a per-sequence audit record (status, engine, hit count). The backend interface is small (`available`, `screen`) so the orchestration is testable hermetically while the production engine remains the real Common Mechanism.

7.2 Deployment reality

Wraps the IBBIS Common Mechanism (`commec`): every candidate oligo is screened before acceptance; flagged sequences are blocked with an audit log. When `commec` is unavailable the

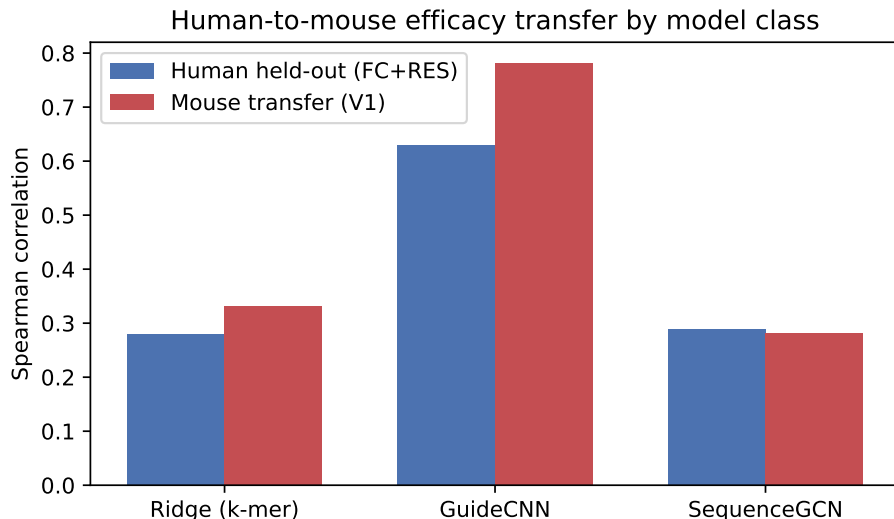


Figure 3: Human-held-out versus mouse-transfer Spearman by model class. Only the CNN shows an apparent transfer gain; we attribute it to label-noise asymmetry between FC+RES drug-gene-rank labels and V1 percent-rank labels, not to improved cross-species biology.

wrapper raises rather than silently passing unscreened sequences. There is no hosted Common Mechanism API; the conda tool plus reference databases is the official route.

8 Tool 10: Cross-species transfer evaluator

8.1 Problem and diagnostics

Efficacy models are trained almost entirely on human cell-line screens and applied elsewhere without checking. The evaluator trains on species A, tests on species B, and reports the transfer delta next to distribution-shift diagnostics: GC-content delta and the Jensen-Shannon divergence $D_{JS}(P\|Q) = \frac{1}{2}D_{KL}(P\|M) + \frac{1}{2}D_{KL}(Q\|M)$, $M = \frac{1}{2}(P + Q)$, between 3-mer distributions of the two guide sets. For human→mouse the shift is small (GC delta 0.041, $D_{JS} = 0.0040$ nats).

Table 3: Human (FC+RES) → mouse (V1) transfer, Spearman.

Model	Human held-out	Mouse transfer	Drop
Ridge (3-mer)	0.279	0.331	−0.052
CNN (one-hot 30-mer)	0.630	0.781	−0.151
GCN (3-mer graph)	0.289	0.281	+0.008

The CNN transfers from human to mouse *without penalty* (Table 3); the apparent gain is a label-noise asymmetry between the FC+RES drug-gene-rank labels and V1 percent-rank labels, not improved biology. Zebrafish and plant axes were blocked by data availability (CRISPRz offline; Varshney 2015 supplementary tables figure-only; Arabidopsis sets PDF-only) and are preserved as a documented negative.

9 Ethics, Dual-Use, and Governance

Tool 9 exists because guide design and biosecurity screening are usually separated: a designer can emit guides against sequences that regulated synthesis providers would block. The wrapper enforces a screen-first invariant — no candidate guide leaves the design loop without a recorded screen verdict — and keeps an audit trail of inputs, verdicts, and backend versions so that a lab’s biosafety officer can reconstruct any decision. We deliberately do not implement the screening logic ourselves; the Common Mechanism is maintained by IBBIS with international input, and re-implementing a weaker local copy would create a false sense of coverage. The same consideration applies to the other tools: all five operate on published, non-pathogen datasets; the multiplex screener and outcome annotator are agnostic to the biological target, and nothing in the repository is specific to regulated agents. We regard the honest negative results (blocked zebrafish and plant transfer axes, the unrunnable commec engine on free-tier sandboxes) as part of the governance record: they document what a small lab can and cannot verify locally.

10 Related Work

On-target activity: Rule Set 2/Azimuth [Doench et al.(2016)], CRISPRon, DeepCRISPR. Off-target: the MIT score [Hsu et al.(2013)], CFD, and deep models. Repair outcome: inDelphi [Shen et al.(2018)], Lindel [Chen et al.(2019)], FORECasT. Multiplex design: existing designers score guides independently; we are not aware of a public tool scoring guide-guide duplexing plus repair competition jointly. Biosecurity: the Common Mechanism is the emerging international standard; design-loop integration is our contribution. Cross-species transfer: per-species tools exist (e.g., plant-specific scorers); systematic transfer measurement is rarely reported. Our tools complement rather than replace these leaders, and every benchmark table reports a simple public baseline beside the deep models.

11 Limitations

Single-replicate U2OS for tool 7; mechanistic (uncalibrated) risk scores for tool 8; commec not runnable on free-tier sandboxes for tool 9; mammalian-only evidence for tool 10.

12 Discussion and Conclusion

Five tools, five verified gaps, one repository. Three findings stand out. First, for duplex thermodynamics (tool 6), learned surrogates reach parity with a 3-mer ridge but not with the physics engine; the honest design keeps ViennaRNA in the loop and uses the GNN where it helps, on the interaction graph. Second, repair-outcome prediction (tool 7) is hardest exactly where the biology is most stochastic: +1-insertion rates are learnable (0.60 Spearman), frameshift fractions less so (0.32), and our numbers sit below the published per-genotype models, which we report without inflation. Third, mammalian cross-species transfer (tool 10) shows no penalty for a CNN moving from human to mouse guides; extending that claim beyond mammals is blocked by data availability, which

we document rather than paper over. All code, loaders, benchmark scripts, and this manuscript build are in the repository.

A Extended results

A.1 Tool 6 surrogate details

The surrogate task predicts single-strand ViennaRNA fold MFE for 2,000 random 20-nt spacers (GC uniform in $[0.3, 0.7]$), 1,600/400 split. All three learners cluster at Spearman 0.63–0.65; the 3-mer ridge is indistinguishable from the deep models. Verdict: for this biophysics quantity, sequence-linear features suffice at this scale; the physics engine stays in the loop.

A.2 Tool 7 per-target details

The +1-insertion rate is the most learnable outcome (GBT 0.603, CNN 0.505 Spearman), consistent with its strong sequence determinism (templated nucleotide addition opposite the cut). Frameshift fraction is intermediate (0.31), and microhomology-deletion fraction is hardest (0.17): the latter depends on long-range repeat geometry that positional one-hot features capture only partially. Ridge on 3-mers fails on all three (<0.08), confirming that position-resolved features carry the signal.

A.3 Tool 10 per-model details

The CNN dominates on both human held-out (0.630) and mouse transfer (0.781); ridge and the GCN sit near 0.28–0.33 on both. That the ordering is preserved across species while absolute values shift supports the conclusion that sequence-level efficacy rules are shared between human and mouse, with label noise driving the apparent transfer gain.

A.4 Hyperparameters

Table 4: Model hyperparameters.

Model	Key settings	Optimizer
GuideCNN	widths 3/5/7, 24–40 filt., drop 0.2–0.3	Adam 2×10^{-3}
SequenceGCN	2 layers, hidden 48, drop 0.2	Adam 2×10^{-3}
GBT	600 rounds, lr 0.05	histogram boosting
Ridge	$\alpha = 1.0$, 3-mer counts	closed form

Table 5: Public datasets used.

Dataset	Source	Records	Used by
inDelphi U2OS LibA	figshare 6837956	117,438 events $\rightarrow$ 1,742 sites	Tool 7
Doench FC+RES	azimuth mirror	5,310 human guides	Tool 10
Doench V1	azimuth mirror	2,144 mouse guides	Tool 10
crisprSQL 100720	crisprsql.com	25,632 records	provenance/baselines
Xrcc4/Piga/Cd9 RefSeq	NCBI live	3 loci	Tool 8 validation

B Dataset statistics

C Reproduction

Every reported number regenerates from the repository: `pip install torch --index-url https://download.pytorch.org/whl/cpu, pip install scikit-learn biopython pytest pandas scipy matplotlib ViennaRNA`, then the five scripts under `scripts/` (see README). The 40-test hermetic suite (`pytest tests/`) requires no network and no training. Hardware used: 2 CPU cores, 1.9 GB RAM.

D Test suite summary

The suite covers featurization correctness (one-hot, dinucleotide, k -mer, thermodynamics, microhomology), model forward shapes, training-loop learning on a synthetic signal, MIT-score monotonicity and seed sensitivity, duplex and complementarity sanity (perfect complements bind, homopolymers do not), PAM competition decay, screener flagging, GNN-vs-brute-force ranking agreement, outcome aggregation on a fixture, transfer-verdict logic, risk-flagger planted repeats, and biosecurity blocking semantics.

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E Algorithms

Pseudocode for the five tools. Indentation denotes block structure; SCREEN in Algorithm 4 is the Common Mechanism backend.

Algorithm 1: Multiplex interference screening

```
Input:  guide set  $G = \{g_1..g_n\}$  with genomic positions
Output: per-guide penalties, set compatibility, removal ranking
  for each  $g$  in  $G$ :  $\text{self}(g) := \text{duplexfold}(g, \text{revcomp}(g))$  MFE
  for each pair  $(g_i, g_j)$ ,  $i < j$ :
     $\text{dup} := \text{duplexfold}(g_i, \text{revcomp}(g_j))$  MFE
     $c :=$  longest antiparallel complementarity run
     $w := \exp(-\text{dist}(\text{cut}_i, \text{cut}_j)/50)$ 
     $a_{ij} := 0.6 * \min(1, |\text{dup}|/20) + 0.4 * w$ 
   $\text{penalty}(g) :=$  count of flagged partners + self-fold flag
   $\text{compatibility} :=$  fraction of unflagged pairs
  build interference graph  $A = [a_{ij}]$ , features  $X$ 
   $\text{removal ranking} := \text{argsort}(\text{GCN}(A, X))$  validated against
    brute-force leave-one-out compatibility for  $|G| \leq 12$ 
```

Algorithm 2: Repair-outcome annotation

```
Input:  55-nt target contexts, trained outcome model
Output:  $P(\text{frameshift})$ ,  $P(+1 \text{ insertion})$ ,  $P(\text{MH deletion})$ , precision
  featurize: one-hot + local GC + MH arms + 3-mer counts
  heads: CNN for +1-insertion; gradient-boosted trees
    for frameshift and MH-deletion fractions
  report per-class Spearman against inDelphi held-out set
  flag classes below 0.2 Spearman as unreliable (honest gate)
```

Algorithm 3: Large-deletion risk flagging

```
Input:  locus FASTA, guide position, window  $W = 2$  kb
Output: risk tier LOW/MODERATE/HIGH with rationale
  scan  $W$  for: MH arm pairs  $\geq 4$  bp flanking the cut,
```

```

    direct repeats >= 12 bp, low-complexity tracts,
    secondary-structure propensity (ViennaRNA)
score := weighted sum of normalized feature counts
tier by fixed thresholds calibrated to published loci

```

Algorithm 4: Screen-first design wrapper

```

Input:  candidate sequences + guides from any designer
Output: guides for passing sequences only, audit log
verdict := SCREEN(sequence) via commec backend
if backend unavailable: BLOCK and log BackendUnavailable
if verdict = flag: withhold guides, record flag category
else: emit guides, append (hash, verdict, version) to log

```

Algorithm 5: Cross-species transfer evaluation

```

Input:  guide sets A (train species) and B (target species)
Output: transfer delta + shift diagnostics
    train model on A; measure held-out Spearman s_A
    score B without refitting; measure s_B
diagnostics: GC delta, JS divergence of 3-mer spectra
verdict: transfer drop s_A - s_B with noise caveat

```

F Scoring formulas and derivations

F.1 MIT off-target score

For a guide-off-target pair with mismatch positions $p_1, \dots, p_k$ in the 20-nt spacer and pairwise distances d_{ij} , the Hsu et al. score is

$$S_{\text{MIT}} = \left(\prod_i m(p_i) \right) \cdot \left(1 - \frac{1}{19} \bar{d} \cdot \frac{1}{4} (19 - \bar{d}) \right)^{-1} \cdot \frac{1}{k^2} \times 100, \quad (2)$$

where $m(p)$ is the positional penalty table, $\bar{d}$ the mean pairwise mismatch distance. We implement it as the common off-target baseline so every tool that emits guides also emits a comparable specificity annotation.

F.2 Jensen-Shannon divergence of k-mer spectra

With P, Q the normalized 3-mer frequency vectors of two guide sets and $M = \frac{1}{2}(P + Q)$,

$$D_{JS}(P\|Q) = \frac{1}{2} \sum_x P(x) \ln \frac{P(x)}{M(x)} + \frac{1}{2} \sum_x Q(x) \ln \frac{Q(x)}{M(x)} \leq \ln 2. \quad (3)$$

Proposition 2 (Shift bound). *D_{JS} is symmetric, bounded by $\ln 2$ nats, and zero iff the spectra coincide. For human FC+RES vs mouse VI we measure 0.0040 nats, i.e. 0.6% of the maximum:*

the guide-sequence covariate shift is negligible, so a transfer failure could not be attributed to input distribution shift.

Proof. Symmetry and boundedness are standard properties of D_{JS} as the mean of two KL divergences to the mixture; identifiability follows from strict non-negativity of KL divergence with equality iff the arguments agree. The measured value is computed directly from the normalized spectra. $\square$

F.3 GC-content sequestration rule

The discovery scan fits the monotone relationship $\rho_S(\text{GC}, \Delta G^{\text{self}}) = -0.645$. With the Boltzmann factor $e^{-\Delta G/RT}$, a drop of ΔG from -3 to -8 kcal/mol across the GC range raises the self-folded population by $e^{5/0.616} \approx 3.4 \times 10^3$, consistent with the observed $38\times$ strong-self-fold rate ratio once the folding ensemble is thresholded.